

Formal Review of the Consolidated v2 Generated Split-Forest Proof and the 2026-05-26 Verification Bundle

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Abstract

This memorandum reviews the latest consolidated manuscript `repaired_split_forest_no_automata_proof_v2_consolidated` and the verification archive `verification-05-26-2026-latest.zip`. The new version is a genuine improvement: the verification bundle is now substantially aligned with the stronger claims, the previous V6 no-mosaic loophole has been closed in the executable checker, strengthened multiple upward-PP tests are present, request-cycle tests are present, and arity-4 mosaic tests are included. The proof architecture is still credible. However, the mathematical proof is not yet closed. The remaining issues are not merely stylistic; several internal inconsistencies remain between the new recursive verticalization discipline, the old side-interface lemma, the support-closure definition, the patchwork proof, and the final enumeration bound. I therefore recommend treating the current manuscript as strong progress, not yet as a finished decidability proof.

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1 Materials reviewed and reproduction checks

I reviewed the following uploaded materials:

- `repaired_split_forest_no_automata_proof_v2_consolidated.pdf`;
- `repaired_split_forest_no_automata_proof_v2_consolidated.tex`;
- `verification-05-26-2026-latest.zip`.

The TeX source compiles locally with `pdflatex` and produces a 36-page PDF. The only compilation warnings I observed after repeated runs were harmless `hyperref` PDF-string warnings.

I also unpacked and reran the verification scripts in `verification/python`. The four main scripts named in the README all exit with code 0:

Script	Exit code	Result
<code>rcc5_composition_check.py</code>	0	PASS
<code>certificate_checker.py</code>	0	PASS
<code>small_model_oracle.py</code>	0	PASS
<code>mosaic_closure_search.py</code>	0	PASS

The additional script `wp7_side_witness_stress.py` is present in the archive, but it does not run from the archive alone: it imports project modules from `src/` that are not included in the zip, yielding `ModuleNotFoundError: No module named 'alcircc5_reasoner'`. This is not a mathematical refutation, but it means that the manuscript's claim that this script gives empirical ground truth is not reproducible from the supplied verification package as distributed.

2 Executive verdict

The latest version is a significant improvement over the previous manuscript. In particular:

- the RCC5 composition table is computationally checked;
- the V6 no-mosaic loophole is fixed in the checker;
- strengthened two-superpart tests, including an incomparability-forcing variant, are present;
- request-closure regressions are present;
- the proof now explicitly attempts recursive verticalization (M6'), abstract pair labels L_Q , overlap cross-labels, joint equality discharge, and uniform cycle representatives.

These are the right repairs. Nevertheless, I would not yet state that the paper establishes decidability. Several proof-critical components are still internally inconsistent or under-proved. The most important remaining defects are:

- (G1) the old side-interface legality lemma still excludes exactly the PP/PPI-comparable side witnesses that M6' was introduced to admit;

- (G2) the support-closure definition was weakened to per-position coverage (SC4'), but the quotient-lifting and patchwork proofs still invoke per-triple mosaic coverage from the old SC4;
- (G3) the overlap/universal-propagation argument does not yet prove that cross-mosaic universal restrictions are preserved;
- (G4) the no-automata regularization section still mixes complete-profile repetition and extended-profile repetition, and the uniform-cycle substitution lemma needs a stronger compatibility proof;
- (G5) the final enumeration bound contradicts the earlier side-width recurrence and checks only V1–V10, although the certificate validity definition now has V1–V14;
- (G6) the verification package is useful evidence but does not prove the support-closed patchwork theorem, and one cited script is not self-contained.

3 Positive progress since the previous round

3.1 The verification bundle is now materially stronger

The earlier mismatch between Claude’s memo and the uploaded verification folder has mostly been corrected. The latest certificate checker contains 26 test cases, including:

- the strengthened C_{AB}^{inc} test;
- the single-parent and single-chain negative controls;
- pure request-cycle positive and negative controls;
- explicit no-mosaic regressions for $\exists \text{DR}.A$ and $\exists \text{PO}.A$;
- occurrence-freshness tests separating blocking from semantic equality.

All of these pass locally. This is valuable counterexample-search evidence.

3.2 Recursive verticalization is the right conceptual repair

The manuscript correctly identifies the round-4 side-witness problem: two DR/PO side witnesses of the same source can themselves be PP/PPI- comparable. The stress concept

$$\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$$

can be satisfied with two DR-witnesses y, z of the source and with $y \text{PP} z$. A flat sibling-frontier restriction to $\{\text{DR}, \text{PO}\}$ is therefore too strong. The proposed recursive verticalization discipline (M6'), where a side witness can open a nested local vertical stratum, is the right direction.

3.3 The executable V6 defect appears fixed

In the previous verification review I found that a certificate for $\exists \text{DR}.A$ with no mosaics was accepted vacuously. The new checker now contains explicit negative tests for **exists DR.A with no mosaics** and **exists PO.A with no mosaics**, both rejected. This is a real improvement.

4 Remaining proof-critical gaps

4.1 The old side-interface lemma contradicts the new M6' discipline

The most immediate mathematical problem is that the consolidated manuscript contains both of the following claims.

First, the new M6' discussion says that two side witnesses y, z of the same source may be PP/PPI-comparable, and that this is handled by opening a nested local vertical stratum. This is the intended round-5 repair.

Second, Lemma **Side-interface legality** still states that the side interface $\text{Side}(u)$ is legal with vertical positions only

$$\{u\} \cup \{\hat{u} : \hat{u} \equiv_{\text{EQ}} u\},$$

and all remaining positions are sibling-frontier positions. It then asserts that labels between sibling-frontier positions lie in

$$\{\text{DR}, \text{PO}, \text{EQ}\},$$

and explicitly excludes the PP/PPI options.

These two statements are incompatible. The stress concept above is exactly a case where two side witnesses of the same source may have $y\text{PP}z$. Under M6' that pair should be co-vertical inside a nested stratum; under the old side-interface lemma it is forbidden. The proof of the lemma even says that a PP-comparable pair of selected witnesses would be “folded into the vertical backbone”, but the definition of $\text{Side}(u)$ does not actually construct the nested stratum that performs this folding.

This is not only a wording problem. Later completeness steps still invoke Lemma **side-placement** to claim that collected side interfaces satisfy (M1)–(M6). If that lemma is the old flat-frontier lemma, it loses satisfiable models. If it is meant to be the new recursive lemma, then its definition and proof must be rewritten.

Required repair. Delete or replace the current Lemma **Side-interface legality**. The replacement should define $\text{Side}(u)$ as a finite *stratified* side network, not as one source/equality cluster plus a flat sibling frontier. It should prove:

- (S1) every PP/PPI label among side positions is co-vertical in some explicitly named nested stratum;
- (S2) cross-stratum labels are DR/PO;
- (S3) the construction has the stated effective width bound;
- (S4) the collected stratified side interfaces satisfy the same M6' used later in the mosaic definition.

4.2 SC4' is per-position, but the patchwork proof still uses per-triple coverage

The manuscript changes support closure to a per-position condition (SC4'): for each abstract position symbol $\pi \in \text{Pos}_Q$, there is a mosaic containing that position with the right type and stratum. It then says that per-triple coverage is not required because V11 enforces the RCC5 composition table on the abstract triple graph.

However, the next lemmas still rely on the old per-triple version of SC4. In particular, Lemma **Quotient lifting** says that every concrete triple (u, v, w) is covered by a mosaic $M_{q(u,v,w)}$ “from (SC4)”, and the proof of Theorem **Support-closed mosaic patchwork** likewise says that every triple of occurrences in a finite prefix is realised in some mosaic by (SC4).

This is an internal contradiction:

SC4' gives mosaics for individual positions,

whereas

the proof uses mosaics for triples.

V11 may be enough to check RCC5 composition on abstract labels, but then the patchwork proof must be rewritten to use L_Q directly, not a triple mosaic that SC4' no longer guarantees.

Required repair. Choose one of two routes and make the whole proof consistent.

Route A: restore triple coverage. Define support closure so that for every abstract triple in Trip_Q there is a mosaic realising that triple. Then Lemma **Quotient lifting** and Theorem **patchwork** can keep their current form, modulo updated M6' stratification.

Route B: remove triple-mosaic dependence. Keep SC4' per-position coverage, but prove a new global-label theorem: every concrete pair label in the unfolding is exactly $L_Q(q(u), q(v))$; every concrete triple therefore satisfies RCC5 composition by V11; and type propagation/existential support are handled by separate V6/V7/V13 clauses. Under this route, Lemma **Quotient lifting** as currently stated must be deleted.

The current manuscript mixes the two routes.

4.3 The abstract triple map is still circular

Definition **Quotient map from concrete to abstract** says that ordered concrete triples are sent to elements of Trip_Q “by (M1)–(M6), (P1)–(P3), and the patchwork lemma.” But the patchwork lemma is later proved using this same quotient-lifting step. This makes the argument circular.

A non-circular proof should first define, independently of patchwork, how every concrete occurrence receives an abstract position symbol and how every concrete pair label is computed from L_Q . Then V11 can be applied directly to concrete triples. Alternatively, if triple mosaics are restored, then $q(u, v, w)$ should be defined from the concrete unfolding and SC4 should provide a mosaic for it, without appealing to the theorem being proved.

4.4 Universal propagation across overlap amalgams is under-proved

Lemma **Universal-propagation preservation** is still too compressed. The hard case is a cross pair $p \in P_1 \setminus P_0$, $q \in P_2 \setminus P_0$ with $L_{12}(p, q) = R$ and $\forall R.D \in \tau(p)$. The proof says that V7 and a composition chain through an overlap point force $D \in \tau(q)$. This is not an immediate consequence of RCC5 composition. Composition determines possible relation labels; it does not by itself propagate type information along a newly declared cross-pair label.

Required repair. The certificate needs an explicit type-safety condition for declared cross-labels: for every declared overlap cross-pair (p, q) with label R , if $\forall R.D \in \tau(p)$, then $D \in \tau(q)$, and symmetrically using inverse labels. Then Lemma **M4-preserved** becomes immediate. Without such a clause, overlap amalgamation can introduce a new R -edge that violates a universal restriction.

4.5 The request-cycle proof still has profile/extended-profile slippage

The request-cycle lemma itself is close to the desired no-automata replacement: it uses half-open cycles and fresh occurrences, and it rejects cycles that carry unfulfilled vertical existential requests. This is the right idea.

There is still a mismatch between the lemmas using it. Lemma **Request-fulfilled repetition** uses repeated *extended* profiles $\hat{\Pi}$, including pending-set information. But Lemma **Lasso construction by blocking** says to take the first repetition of the ordinary complete profile $\Pi(u_i) = \Pi(u_j)$ and then invokes the request-cycle lemma. That is not sufficient: two occurrences can have the same complete boundary profile but different pending request sets. The lasso lemma should repeat $\hat{\Pi}$, not just Π .

There is also a smaller half-open notation issue. The proof later says to replace the tail by fresh copies of $[u_i, u_j]$, while the convention defines the cycle word as $[u_i, u_j)$. This should be made consistent throughout.

Required repair. State the lasso lemma only for repeated extended profiles:

$$\hat{\Pi}(u_i) = \hat{\Pi}(u_j),$$

and use the half-open segment $[u_i, u_j)$ everywhere. Then define the finite cycle validity check to verify request fulfillment on that half-open word.

4.6 Uniform global regularization needs a stronger compatibility lemma

Lemma **Uniform global regularization** says that all rays hitting the same extended profile can use one representative cycle word. This is plausible only if the extended profile records every boundary component needed for substitution: context scheme, port pattern, side mosaic interfaces, M6' stratum forest, equality cluster, request status, and every outward relation summary. The manuscript asserts this through the substitution lemma, but the proof does not show that the representative cycle is compatible with every occurrence of that extended profile in every ray.

This is another place where the proof should be made more mechanical. The complete profile should be defined as a finite boundary isomorphism type, and the substitution lemma should explicitly say that equal complete profiles induce an isomorphism of all boundary data, including the stratum forest and all named mosaic positions. Then the uniform-cycle lemma can rely on that isomorphism.

4.7 The side-width and enumeration bounds are inconsistent

The manuscript first says that recursive verticalization requires a side-width recurrence over modal depth:

$$m(0) = 1, \quad m(d+1) = 1 + |\text{Ex}_{\text{DR/PO}}(u)|(1 + k_{\text{Mate}} + k_{\text{bd}}) + |\text{Ex}_{\text{DR/PO}}(u)|m(d),$$

and explicitly states that this is not polynomial in general.

Later, in the decision-procedure section, the manuscript reverts to a cubic bound

$$m(n) = 1 + n^3 + 3n^2 + 3n = O(n^3)$$

for every generated occurrence. This contradicts the earlier recurrence. The abstract and the enumeration bound also speak of a computable double-exponential bound, while the final proof of Lemma **enumeration** says that the certificate validity clauses V1–V10 are checked, omitting the newly added V11–V14.

Required repair. Use one bound consistently. The safer bound is the recurrence-derived computable bound, for example

$$m_*(C_0) = m(\text{md}(C_0)).$$

Then define the certificate grammar and enumeration bound using $m_*(C_0)$, not an unrelated cubic $m(n)$. Also update every decision-procedure statement from V1–V10 to V1–V14.

4.8 The certificate grammar has not caught up with M6'

The mosaic definition now uses a stratum forest

$$\mathcal{V} = (\text{Stratum}, \text{Subsrc}),$$

but several later pieces still treat the vertical component as a simple subset $\text{Vert} \subseteq P$. Examples include the grammar entry

$$\text{Stratum } \text{Vert}_M \subseteq P$$

and lemmas that write mosaics as $(P, \tau, L, \text{Vert})$ rather than $(P, \tau, L, \mathcal{V})$. This is not merely notation: a subset Vert cannot encode nested strata, parent-position maps, or the co-vertical relation required by M6'.

Required repair. Update the grammar, the finite alphabet count, and every mosaic lemma to use the full finite stratum-forest object. The enumeration bound must count finite stratum forests of size at most the side-width bound.

5 Assessment of the verification bundle

The verification evidence is useful and has improved. It now supports several finite aspects of the proof architecture:

- the RCC5 composition table is correctly implemented;
- the strengthened multiple-superpart examples are included;
- occurrence-level equality is distinguished from blocking cycles;
- no-mosaic DR/PO witnesses are rejected;
- arity-4 atomic mosaic networks and overlap amalgamations are tested.

However, the verification bundle remains counterexample-search evidence, not a formal proof. In particular:

1. The arity-4 mosaic search does not prove the full support-closed patchwork theorem for arbitrary finite prefixes.
2. The checker does not appear to test the new SC4' versus triple-coverage inconsistency identified in Section 4.2.
3. The side-witness stress script is not self-contained in the supplied archive, because it depends on missing `src/` modules.

4. The small-model oracle is useful for finite sanity checks, but it cannot validate infinite regular witnesses such as $\exists PP.\top \sqcap \forall PP.\exists PP.\top$ beyond confirming that no small finite model exists.

I recommend including either the full project source needed by `wp7_side_witness_stress.py`, or replacing that script with a self-contained finite model construction for the stress concept. The latter would be easy: take three regions/elements x, y, z with $xDRy$, $xDRz$, and $yPPz$, put A on y and B on z , then verify the concept directly.

6 Recommended next revision

The next revision should prioritize coherence over adding new machinery. I recommend the following concrete edit plan.

- (R1) Replace the old **Side-interface legality** lemma with a genuinely recursive $M6'$ side-interface construction. Do not use the old flat sibling-frontier lemma anywhere in completeness.
- (R2) Decide whether support closure is per-triple or per-position. If per-position, rewrite the patchwork proof to use L_Q directly and remove Lemma **Quotient lifting**. If per-triple, restore SC4 as triple coverage.
- (R3) Add an explicit cross-pair universal-safety clause for overlap cross-labels, and use it to prove Lemma **M4-preserved**.
- (R4) Change the lasso construction to repeat extended profiles $\hat{\Pi}$, and standardize all cycle words as half-open $[i, j)$ segments.
- (R5) Define the complete profile as a finite boundary isomorphism type that includes the full stratum forest, all ports, all side mosaic positions, and all request statuses. Then restate the substitution lemma in terms of that boundary isomorphism.
- (R6) Replace the cubic side-width bound by the recurrence-derived computable bound $m(\text{md}(C_0))$, or prove that recursive verticalization can actually be compressed to the cubic bound. At present the two claims conflict.
- (R7) Update the certificate grammar and enumeration proof to cover V1–V14 and to count full stratum forests, not only subsets $\text{Vert} \subseteq P$.
- (R8) Make `wp7_side_witness_stress.py` reproducible from the supplied zip, or remove manuscript claims that rely on it.

7 Final assessment

The proof has moved in the right direction. I am more confident now than in the previous round that the intended split-forest architecture can be made to work: witness thinning, generated containment covers, occurrence-level equality ports, recursive side verticalization, support-closed mosaics, and request-fulfilled blocking cycles are the right ingredients.

But the current consolidated manuscript still does not, as written, establish decidability. The largest remaining issue is no longer the original density or Hasse-diagram concern. It is the mismatch among three layers that now have to be aligned precisely:

recursive side strata ($M6'$), abstract pair/triple support, finite regular certificate enumeration.

Once those layers are made internally consistent, the proof will be much closer to publication-grade. Until then, the appropriate verdict is:

Substantial progress, but the decidability proof remains open as a formal manuscript.
